

Agentic Frameworks for Runtime Self-Correction: Comprehensive Guide

Frontend Technologies

- **React** 18+ with **TypeScript**
- **Next.js** 14 for SSR and API routes
- Tailwind CSS for styling
- Redux Toolkit for state management
- React Query for data fetching
- D3.js for visualization
- WebSocket API for real-time updates

Backend Technologies

- **Node.js** with Express.js
- Python FastAPI for ML services
- GraphQL with Apollo Server
- PostgreSQL for primary database
- Redis for caching and sessions
- MongoDB for unstructured data
- Docker for containerization
- Kubernetes for orchestration

AI/ML Stack

- **PyTorch**/TensorFlow for deep learning
- Hugging Face Transformers
- LangChain for LLM orchestration
- OpenAI API / Anthropic Claude
- Vector databases (Pinecone/Chroma)
- MLflow for experiment tracking
- Apache Airflow for ML pipelines

Infrastructure & DevOps

- AWS/GCP cloud platforms
- Terraform for infrastructure as code
- GitHub Actions for CI/CD
- Monitoring: Prometheus + Grafana
- Logging: ELK Stack
- Security: OAuth 2.0, JWT tokens